Lab 3: Proof

Answer the following questions given in the book " Introduction to the Theory of Computation by Michael Sipser". (Pages 159-161)

Note: Although the solutions are given in the book, you are required to give the detailed solution by yourself. Copying the solution in the book will lead to zero marks.

[A]**3.3** Modify the proof of Theorem 3.16 to obtain Corollary 3.19, showing that a language is decidable iff some nondeterministic Turing machine decides it. (You may assume the following theorem about trees. If every node in a tree has finitely many children and every branch of the tree has finitely many nodes, the tree itself has finitely many nodes.)

THEOREM 3.16 ....

Every nondeterministic Turing machine has an equivalent deterministic Turing machine.

COROLLARY 3.19 ....

A language is decidable if and only if some nondeterministic Turing machine decides it.